



TRIBHUVAN UNIVERSITY (12, Bold, Center Justified)
INSTITUTE OF ENGINEERING
THAPATHALI CAMPUS

A Project Report (12, Bold, Center Justified)
On
[Project Title]

Submitted By: (12, Bold, Center Justified)
[Student Name] ([Student Exam Roll No.]) (14, Center Justified)
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Submitted To: (12, Bold, Center Justified)
Department of Electronics and Computer Engineering
Thapathali Campus (12, Center Justified)
Kathmandu, Nepal

July, 2019 (12, Center Justified)



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[Student Name] ([Student Exam Roll No.]) (12, Center Justified)

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[Student Name] ([Student Exam Roll No.])

Submitted To: (12, Bold, Center Justified)

Department of Electronics and Computer Engineering

Thapathali Campus

Kathmandu, Nepal (12, Center Justified)

In partial fulfillment for the award of the Bachelor's Degree in Electronics and
Communication Engineering. (12, Center Justified)

Under the Supervision of (12, Bold, Center Justified)

[Supervisor's Name] (12, Center Justified)

July, 2019 (12, Center Justified)

DECLARATION

We hereby declare that the report of the project entitled “**Project Title**” which is being submitted to the **Department of Electronics and Computer Engineering, IOE, Thapathali Campus**, in the partial fulfillment of the requirements for the award of the Degree of Bachelor of Engineering in **Electronics and Communication Engineering**, is a bonafide report of the work carried out by us. The materials contained in this report have not been submitted to any University or Institution for the award of any degree and we are the only author of this complete work and no sources other than the listed here have been used in this work.

Bimal Aryal (Class Roll No: 070/BEX/101) _____

[Student Name] (Class Roll No: [070/BEX/101]) _____

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Date: July, 2019

CERTIFICATE OF APPROVAL

The undersigned certify that they have read and recommended to the **Department of Electronics and Computer Engineering, IOE, Thapathali Campus**, a minor project work entitled “**Project Title**” submitted by **Student Name, Student Name, Student Name** and **Student Name** in partial fulfillment for the award of Bachelor’s Degree in Electronics and Communication Engineering. The Project was carried out under special supervision and within the time frame prescribed by the syllabus.

We found the students to be hardworking, skilled and ready to undertake any related work to their field of study and hence we recommend the award of partial fulfillment of Bachelor’s degree of Electronics and Communication Engineering.

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Department of Electronics and Computer Engineering, Thapathali Campus

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July, 2019

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ACKNOWLEDGEMENT

Your acknowledgement here. Your acknowledgement here. Your acknowledgement here. Your acknowledgement here. **(1 – 5 paragraphs)**

Sujan Dhungana (Class Roll No.: 070/BEX/142)

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ABSTRACT

Your abstract here. Your abstract here. Your abstract here. Your abstract here. Your abstract here. Your abstract here. Your abstract here. Your abstract here. Your abstract here. **(a paragraph)**

Keywords: [most frequently used terms in your report (may be technology, algorithms etc.) (Italic, alphabetical order)]

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List of Abbreviations

AC	Alternating Current
ADC	Analog to Digital Converter
API	Application Programming Interface
BCM	Broadcom
CMOS	Complementary Metallic Oxide Semiconductor
CMS	Center Monitoring System
CSS	Cascading Style Sheets
DHCP	Dynamic Host Control Protocol
et al.	And Others
FTP	File Transfer Protocol
FTPS	File Transfer Protocol Secure
GIS	Geographic Information System
GPIO	General Purpose Input Output
GPRS	General Packet Radio Service
GPS	Global Positioning System
GPU	Graphics Processor Unit
GUI	Graphical User Interface
HDMI	High Definition Media Interface
HTTP	Hypertext Transfer Protocol
HTML	Hypertext Markup Language
IDLE	Integrated Development Environment
IP	Internet Protocol
IOT	Internet Of Things
ISP	Internet Service Provider
ISR	Interrupt Service Routine
IT	Information Technology

1. INTRODUCTION

Your introduction here. Your introduction here. Your introduction here. Your introduction here. Your introduction here. Your introduction here. Your introduction here.

1.1 Background

Your introduction here. Your introduction here. Your introduction here. Your introduction here.

1.2 Motivation

Your introduction here. Your introduction here. Your introduction here. Your introduction here.

1.3 Problem Definition

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1.4 Project Objectives

The main objectives of our project are listed below **(maximum 2 points and to the point):**

- point 1
- point 2

1.5 Project Scope and Applications

Your introduction here. Your introduction here. Your introduction here. Your introduction here.

1.6 Report Organization

Briefly explains all the chapters and their focus. Your introduction here. Your introduction here. Your introduction here. Your introduction here. Your introduction here. Your introduction here. Your introduction here. Your introduction here.

2. LITERATURE REVIEW

This chapter contains all the existing works that have already been carried out in the field related to your project topic. This chapters tells how much you researched before completing your project. You have to explain each of the works as a separate sub-topic with following details:

- **What is the work of existing/researched related topic?**
- **How it is done? used methods, techniques, technology, algorithms and any new innovations of existing/researched related topic)**
- **Its importance or applications existing/researched related topic**
- **drawbacks and limitations existing/researched related topic**
- **Criticize the work of existing/researched related topic**
- **Link these criticisms on the existing/researched related topic to the motivation explained in previous chapter.**
- **Each information should be properly sited.**

Your literature review here. Your literature review here. Your literature review here.
Your literature review here. Your literature review here. Your literature review here.
Your literature review here. Your literature review here [1].

2.1 Sub-heading 1

Your literature review here. Your literature review here. Your literature review here.
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Your literature review here. Your literature review here [2].

2.2 Sub-heading 2 and so on

Your literature review here. Your literature review here. Your literature review here.
Your literature review here. Your literature review here. Your literature review here.
Your literature review here. Your literature review here [3].

3. REQUIREMENT ANALYSIS

Enumerate & describe why and where in your project the hardware components / instruments & software are required. Description of requirement analysis. Description of requirement analysis. Description of requirement analysis. Description of requirement analysis [4].

3.1 Sub-heading 1 – may be Hardware/Software Requirements

Description of requirement analysis. Description of requirement analysis. Description of requirement analysis. Description of requirement analysis. Description of requirement analysis.

3.2 Sub-heading 2 – may be Dataset analysis

Description of requirement analysis. Description of requirement analysis. Description of requirement analysis. Description of requirement analysis. Description of requirement analysis

4. SYSTEM ARCHITECTURE AND METHODOLOGY

Explain the system block diagram along with the purpose of each block & how the blocks interact with each other. Explain any flowcharts / algorithms / methods that your project follows. Your system and architecture here. Your system and architecture here. Your system and architecture here. Your system and architecture here. Your system and architecture here.

Center Justified Figure

Figure 4-1: Block Diagram of abc

4.1 Sub-heading 1 – Block Diagram/Architecture

(Explain all the building blocks of your system in details explaining what and how it does the things) Your system and architecture here. Your system and architecture here. Your system and architecture here. Your system and architecture here. Your system and architecture here.

Center Justified Figure

Figure 4-2: Flowchart of xyz

4.2 Sub-heading 2 – Flowcharts/Algorithms (and other design methods)

Your system and architecture here. Your system and architecture here. Your system and architecture here. Your system and architecture here. Your system and architecture here.

5. IMPLEMENTATION DETAILS

Describe how the hardware components / instruments & software function in your project. Describe the calibration process required for correct operation of each module. Describe the interfacing technicalities / protocol of each module used in your project. Explain in detail how all components are interconnected to make a functioning system. Your system components description here. Your system components description here. Your system components description here. Your system components description here.

5.1 Sub-heading 1 for e.g. Hardware Components

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5.2 Sub-heading 2

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5.2.1 Sub-heading of 5.2

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6. RESULTS AND ANALYSIS

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6.1 Sub-heading 1

6.1.1 Sub-heading 1 of 6.1

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Center justified Figure

Figure 6-1: figure xyz

6.2 Sub-heading 2

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7. CONCLUSION

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8. APPENDICES

It may contains the additional topics or data sheets or reference sheets or even user manual. **Project Budget (Detailed Breakdown of Costs), Project Timeline (Gantt chart), Circuit Diagrams (Should be Clear and Legible), PCB Designs (Should be Clear and Legible), Module Specifications (Should be brief - Keep only necessary tables and figures), Details of Dataset can be included here.** The appendix name should be given as follows.

Appendix A: Circuit Diagram

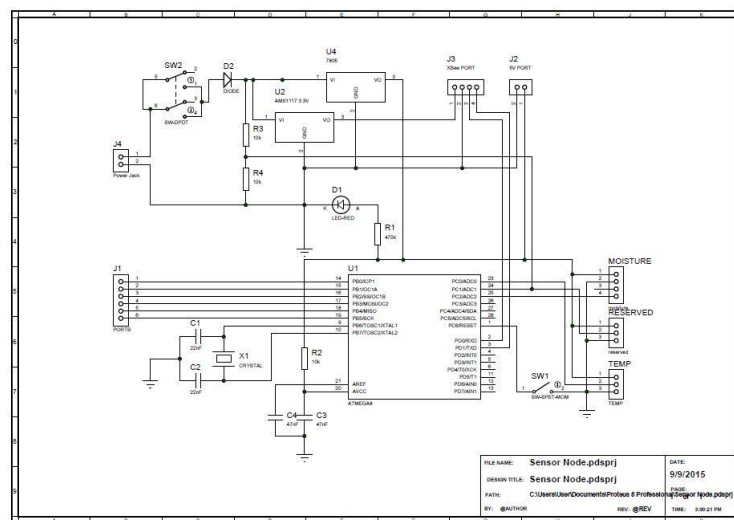


Figure 8-1: Circuit Diagram of AVR with sensors and XBee

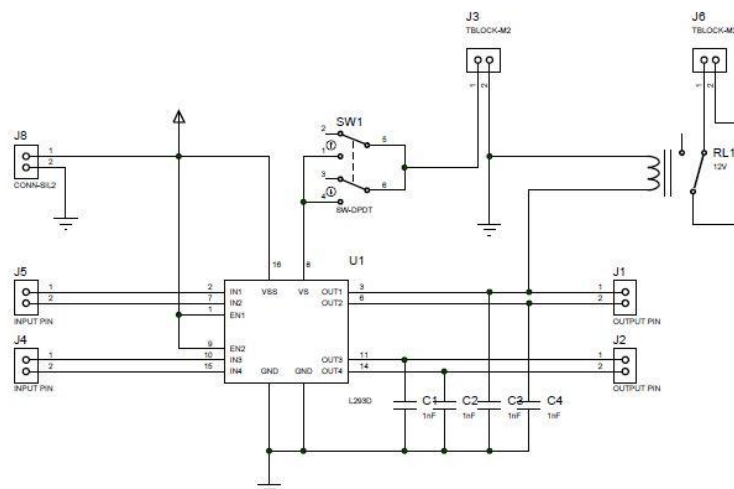


Figure 8-2: Circuit Diagram of Motor Driver and Relay

Appendix B: PCB Diagram

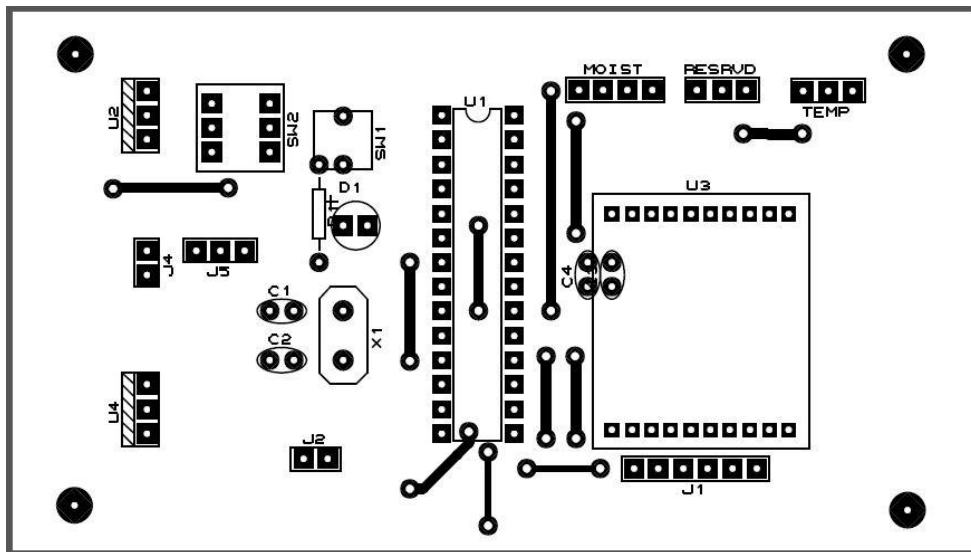


Figure 8-3: PCB Diagram of AVR with sensors and Xbee (Top Layer)

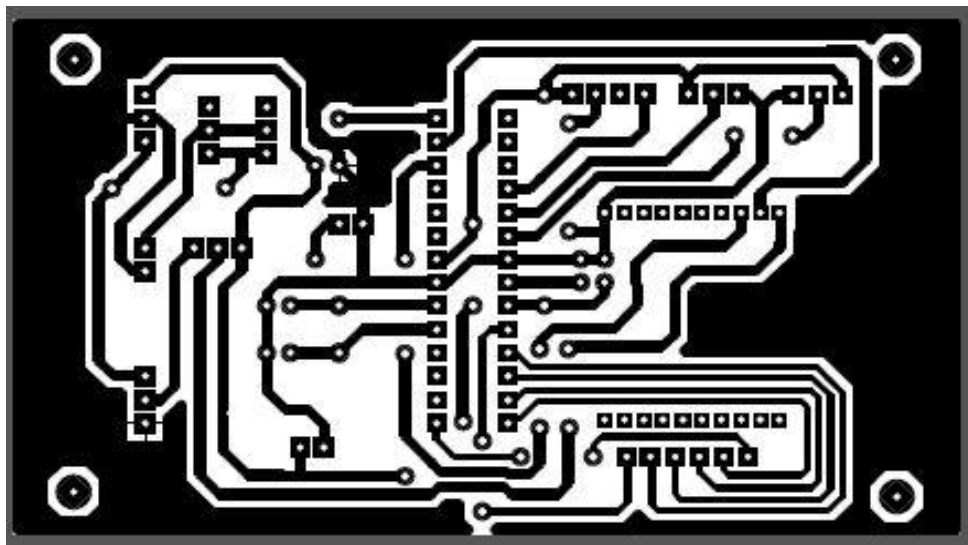


Figure 8-4: PCB Diagram of AVR with sensors and Xbee (Bottom Layer)

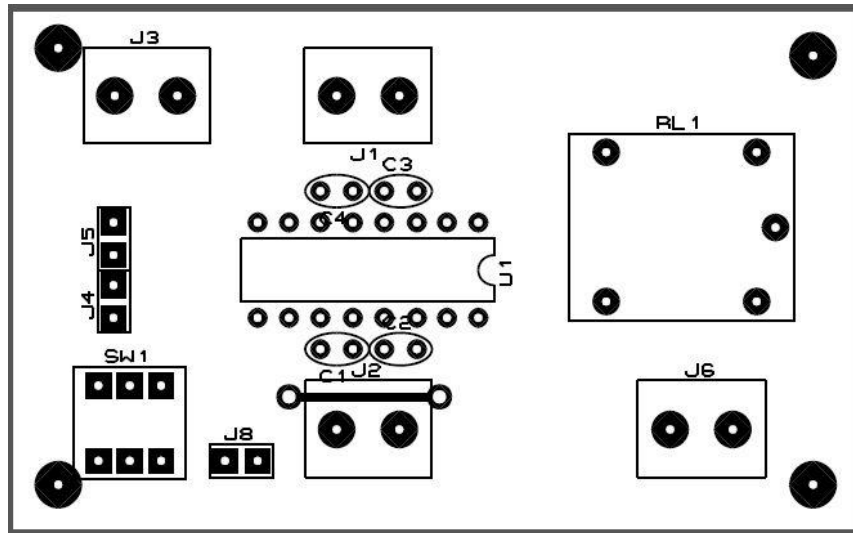


Figure 8-5: PCB Diagram of Motor Driver and Relay (Top Layer)

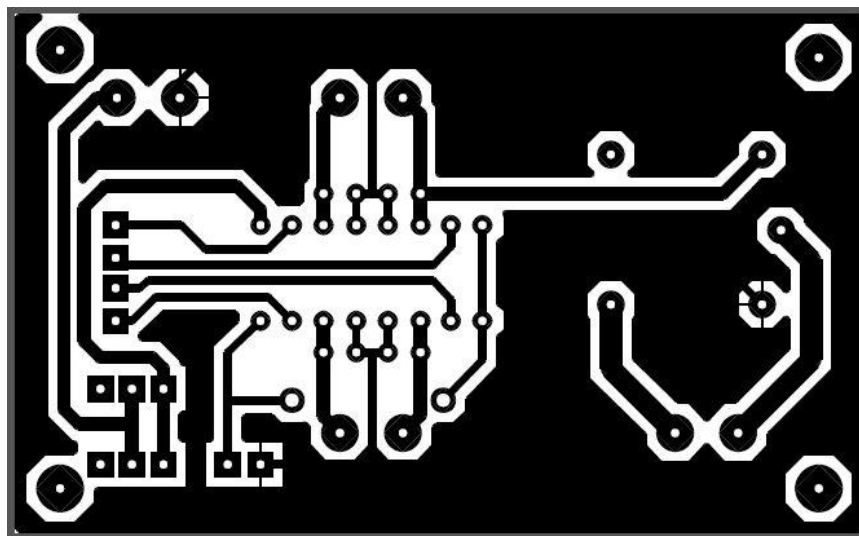


Figure 8-6: PCB Diagram of Motor Driver and Relay (Bottom Layer)

Appendix C: Zigbee Module Specification

Table 8-1: XBee S2 Pin Description [4]

Pin #	Name	Direction	Description
1	VCC	-	Power supply
2	DOUT	Output	UART Data Out
3	DIN / CONFIG	Input	UART Data In
4	DO8*	Output	Digital Output 8
5	RESET	Input	Module Reset (reset pulse must be at least 200 ns)
6	PWM0 / RSSI	Output	PWM Output 0 / RX Signal Strength Indicator
7	PWM1	Output	PWM Output 1
8	[reserved]	-	Do not connect
9	DTR / SLEEP_RQ / DI8	Input	Pin Sleep Control Line or Digital Input 8
10	GND	-	Ground
11	AD4 / DIO4	Either	Analog Input 4 or Digital I/O 4
12	CTS / DIO7	Either	Clear-to-Send Flow Control or Digital I/O 7
13	ON / SLEEP	Output	Module Status Indicator
14	VREF	Input	Voltage Reference for A/D Inputs
15	Associate / AD5 / DIO5	Either	Associated Indicator, Analog Input 5 or Digital I/O 5
16	RTS / AD6 / DIO6	Either	Request-to-Send Flow Control, Analog Input 6 or Digital I/O 6
17	AD3 / DIO3	Either	Analog Input 3 or Digital I/O 3
18	AD2 / DIO2	Either	Analog Input 2 or Digital I/O 2
19	AD1 / DIO1	Either	Analog Input 1 or Digital I/O 1
20	AD0 / DIO0	Either	Analog Input 0 or Digital I/O 0

* Function is not supported at the time of this release

Table 8-2: XBee S2 General Features [4]

Specification	XBee
Performance	
Indoor/Urban Range	Up to 100 ft (30 m)
Outdoor RF line-of-sight Range	Up to 300 ft (90 m)
Transmit Power Output (software selectable)	1mW (0 dBm)
RF Data Rate	250,000 bps
Serial Interface Data Rate (software selectable)	1200 bps - 250 kbps (non-standard baud rates also supported)
Receiver Sensitivity	-92 dBm (1% packet error rate)
Power Requirements	
Supply Voltage	2.8 – 3.4 V
Transmit Current (typical)	45mA (@ 3.3 V)
Idle / Receive Current (typical)	50mA (@ 3.3 V)
Power-down Current	< 10 μ A
General	
Operating Frequency	ISM 2.4 GHz
Dimensions	0.960" x 1.087" (2.438cm x 2.761cm)
Operating Temperature	-40 to 85° C (industrial)
Antenna Options	Integrated Whip, Chip or U.FL Connector, RPSMA Connector

Appendix D: Used Linux Commands [3]

General Commands

- *apt-get update*: Updates your version of Raspbian.
- *apt-get upgrade*: Upgrades all of the software packages you have installed.
- *clear*: Clears the terminal screen of previously run commands and text.
- *Date*: Prints the current date.
- *date -s "3 Sep 2015 10:18:00"*: Reset time and date
- */etc/init.d/ntp start*: restart ntp.
- *find / -name example.txt*: Searches the whole system for the file example.txt and outputs a list of all directories that contain the file.
- *nano example.txt*: Opens the file example.txt in “Nano”, the Linux text editor.
- *poweroff*: To shutdown immediately.
- *raspi-config*: Opens the configuration settings menu.
- *reboot*: To reboot immediately.
- *shutdown -h now*: To shutdown immediately.
- *shutdown -h 01:22*: To shutdown at 1:22 AM.
- *startx*: Opens the GUI (Graphical User Interface).

File/Directory Commands

- `cat example.txt`: Displays the contents of the file `example.txt`.
- `cd /abc/xyz`: Changes the current directory to the `/abc/xyz` directory.
- `cp XXX`: Copies the file or directory `XXX` and pastes it to a specified location; i.e. `cp examplefile.txt /home/pi/office/` copies `examplefile.txt` in the current directory and pastes it into the `/home/pi/` directory. If the file is not in the current directory, add the path of the file's location (i.e. `cp /home/pi/documents/examplefile.txt /home/pi/office/` copies the file from the documents directory to the office directory).
- `ls -l`: Lists files in the current directory, along with file size, date modified, and permissions.
- `mkdir example_directory`: Creates a new directory named `example_directory` inside the current directory.
- `mv XXX`: Moves the file or directory named `XXX` to a specified location. For example, `mv examplefile.txt /home/pi/office/` moves `examplefile.txt` in the current directory to the `/home/pi/office` directory. If the file is not in the current directory, add the path of the file's location (i.e. `cp /home/pi/documents/examplefile.txt /home/pi/office/` moves the file from the documents directory to the office directory). This command can also be used to rename files (but only within the same directory). For example, `mv examplefile.txt newfile.txt` renames `examplefile.txt` to `newfile.txt`, and keeps it in the same directory.
- `rm example.txt`: Deletes the file `example.txt`.
- `rmdir example_directory`: Deletes the directory `example_directory` (only if it is empty).
- `scp user@10.0.0.32:/some/path/file.txt`: Copies a file over SSH. Can be used to download a file from a desktop/laptop to the Raspberry Pi.
- `touch`: Creates a new, empty file in the current directory.

Networking/Internet Commands

- *ifconfig*: To check the status of the wireless connection you are using (to see if wlan0 has acquired an IP address).
- *iwconfig*: To check which network the wireless adapter is using.
- *iwlist wlan0 scan*: Prints a list of the currently available wireless networks.
- *iwlist wlan0 scan | grep ESSID*: Use grep along with the name of a field to list only the fields you need (for example to just list the ESSIDs).
- *nmap*: Scans your network and lists connected devices, port number, protocol, state (open or closed) operating system, MAC addresses, and other information.
- *ping*: Tests connectivity between two devices connected on a network. For example, ping 10.0.0.32 will send a packet to the device at IP 10.0.0.32 and wait for a response. It also works with website addresses.
- *wget http://www.website.com/example.txt*: Downloads the file example.txt from the web and saves it to the current directory.

System Information Commands

- *cat /proc/meminfo*: Shows details about your memory.
- *cat /proc/partitions*: Shows the size and number of partitions on your SD card or hard drive.
- *cat /proc/version*: Shows you which version of the Raspberry Pi you are using.
- *df -h*: Shows information about the available disk space.
- *df /*: Shows how much free disk space is available.
- *dpkg --get-selections | grep XXX*: Shows all of the installed packages that are related to XXX.
- *dpkg --get-selections*: Shows all of your installed packages.
- *free*: Shows how much free memory is available.
- *hostname -I*: Shows the IP address of your Raspberry Pi.
- *lsusb*: Lists USB hardware connected to your Raspberry Pi.
- *vcgencmd measure_temp*: Shows the temperature of the CPU.
- *vcgencmd get_mem arm && vcgencmd get_mem gpu*: Shows the memory split between the CPU and GPU.

References

(Use IEEE format as follows. Note: This is auto generated reference, do not try to type manually. Should be left justified.)

- [1] B. A., "Wireless Sensor Networks in Precision Agriculture," June 2005.

- [2] C. Ayday and S. Safak, "Application of Wireless Sensor Networks with GIS on the Soil Moisture Distribution Mapping," January 2009.

- [3] Circuitbasics.com, "Useful Raspberry Pi Commands," [Online]. Available: <http://www.circuitbasics.com/useful-raspberry-pi-commands/>. [Accessed Aug 2015].

- [4] Digi.com, "Documentation for XBee ZB (S2B) Modules," [Online]. Available: http://ftp1.digi.com/support/documentation/90000976_W.pdf. [Accessed Jun 2015].

- [5] P. Wood, "Raspberry Pi and Custard for Schools," [Online]. Available: <http://www.rs-online.com/designspark/electronics/blog/raspberry-pi-and-custard-for-schools>. [Accessed 02 Sept 2015].

- [6] B. Corporation, "Broadcom BCM2835 ARM Peripherals," Feb 2012. [Online]. Available: <http://www.farnell.com/datasheets/1521578.pdf>. [Accessed 2015 Jun 05].

- [7] Raspberrypi.org, "GPIO: RASPBERRY PI MODELS A AND B," [Online]. Available: <https://www.raspberrypi.org/documentation/usage/gpio/>. [Accessed Jun 2015].

- [8] J. E. K. B. a. S. K. Vongsagon Boonsawat, "XBee Wireless Sensor Networks for Temperature," 2010.

GUIDELINES

- Language must be English Language.
- You must use “**Times New Roman**” font with “**Bold**” and “*Italic*” variants whenever needs (except in exported image/diagrams, screenshot of tables etc.).
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- You can start the numbering from title page but **must not number the cover page.**

- For extensive table of contents and list of figures, you can use 1 line spacing instead of 1.5 (**For extensive Table of contents and list of figures only, otherwise use 1.5 line spacing as normal paragraph**)
- All the figures, tables, charts, equations etc. used in the reports must be numbered using insert caption under reference of MS Word.
- Reference must be used in ascending order (It means if you use [1] reference in page 5 and reference [2] in page 6 then you must not use reference [3] in page 4) using IEEE format.
- Reference should not be numbered as chapter.